

- 4      (a)      Given that  $y = (x-1)\sqrt{4x+1}$ , show that  $\frac{dy}{dx}$  can be written in the form  $\frac{px+q}{\sqrt{4x+1}}$  where  $p$  and  $q$  are constants. [4]

- (b)      Hence evaluate  $\int_2^6 \frac{x}{\sqrt{4x+1}} dx$  [5]